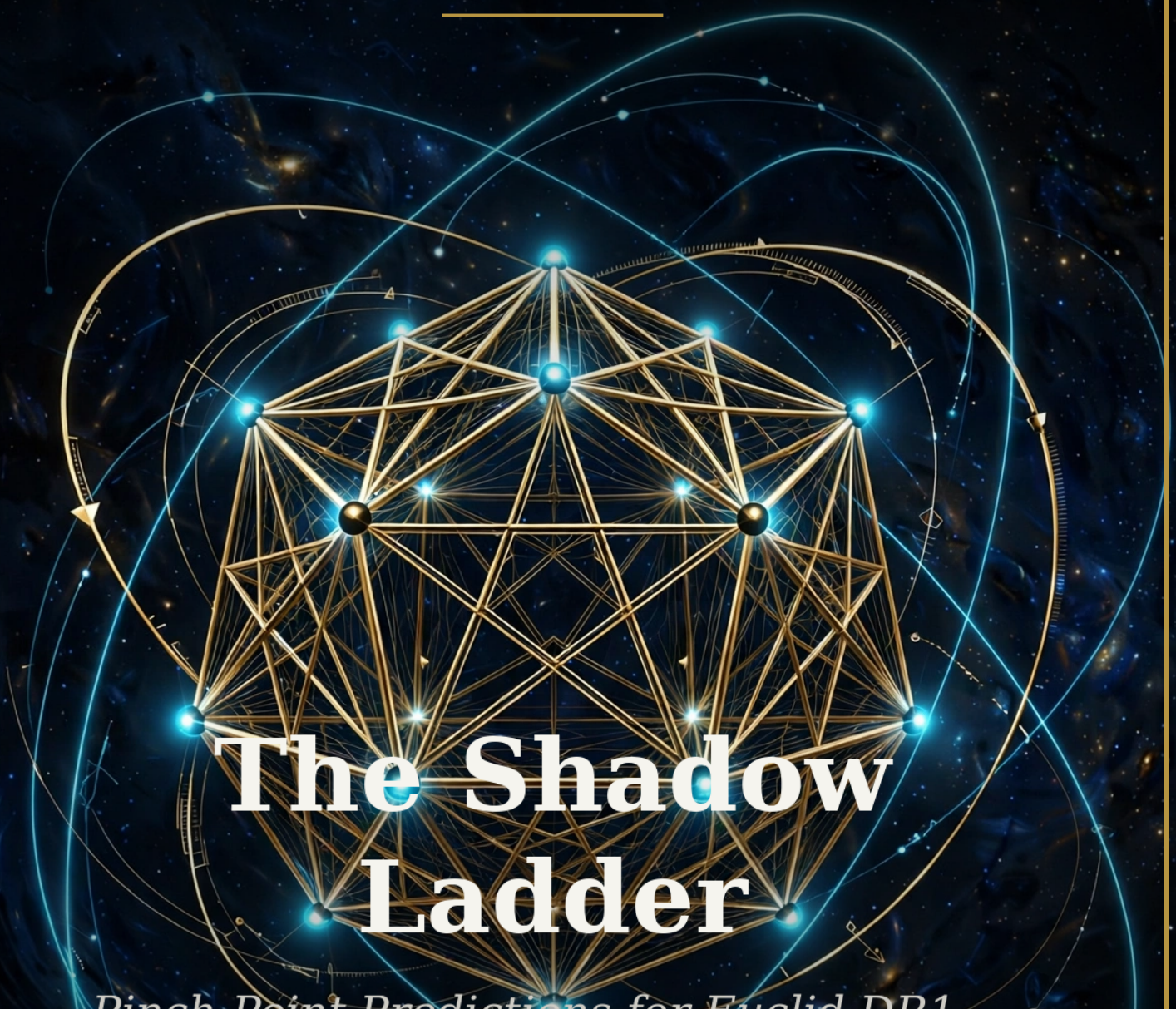
GEOMETRICALLY ORDERED DYNAMICS
FOLIO IT-SHADOW



The Shadow Ladder

Pinch-Point Predictions for Euclid DR1 — Locked Predictions, Verdict November 12, 2026

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RESULTS STATED IN TIER — LIMITS FIRST

CONVERGENCE -- THE SHADOW (E8xE8)

This folio converges on **the Shadow** as the boundary between the visible and shadow sectors of the $E_8 \times E_8$ heterotic structure. Its unique angle is the **pre-registered pinch-point predictions for Euclid DR1**: a 12.34% dipole asymmetry in the weak-lensing signal and ten caustic vertex targets with sky coordinates, computed before any DR1 data product existed, with explicit kill criteria stated in §4. Sibling folios approach the same claim from other angles: the thirteen successor folios derive the framework's sectors, operators, and observables. Limits first: the S/N figures carry the [provisional forecast] tag (§8), and the program's own triangulation gate has already FAILED — recorded here as a failure, not a near-win.

SUPERSESSION NOTE

This paper is the archival freeze of the pre-registration posted 2026-10-12 (Block BLK-119). It supersedes the 2026-10-13 draft, which is retained separately as a superseded record, not a version, per the canon rule: one correct paper, one canon — the correction *becomes* the paper. Honest rewrite 2026-10-30 (six editorial edits per the 2026-10-29 resurrection scoping; zero recomputation, no numbers changed); $E_8 \times E_8$ clarification added 2026-11-01.

Abstract

The Shadow Ladder is a geometric program: the exceptional Lie group E_8 is taken as the generating object ("the exceptional object"), 4D spacetime as its projected shadow, and the projection angle as something to be *derived* from measured mixing angles rather than guessed. On 2026-10-12 the program closed two of its open seeds: (i) the "hexagonal crystal inside the dodecahedron" was resolved — the hexagon is the shadow of an inscribed Kepler cube, via Kepler, *Harmonices Mundi* (1619), verified link-by-link from scratch; (ii) the first Euclid-checkable predictions were computed — a 12.34% dipole asymmetry in the weak-lensing signal across Euclid's footprint (forecast 123.9σ at DR1 footprint / 293.2σ at full survey [provisional forecast — see §8], crew computation, audit of the filed metrics in §3) and 10 pre-registered caustic vertex targets with sky coordinates. Both shape fits are published (dodecahedron residual 6.48°, hexagonal-prism residual 8.71°) so DR1 judges between them. Explicit kill criteria are stated in §4, including the program's own failed gates.

1 The framework in brief

The ladder, bottom to top: **2D screen → 3D hologram → 4D caster → E_8 source.** Spacetime is the shadow; the object is E_8; the light is symmetry-breaking at an angle; the wall is 4D spacetime. The program's standing rule: *do not guess the projection angle — derive it from measured mixing angles* (Weinberg ∼28.7°, Cabibbo ∼13°, PMNS). The 17.5° hypothesis is now parameter-anchored: the settled constant $\kappa = 0.9539$ is the corpus value $\sqrt{0.91} = 0.953939$ ($\theta(2-\theta)$ at $\theta = 7/10$ — the χ of the two-channel "at least one" window; Lean-proved `or_window_mass`). Correction 2026-10-13 ∼15:30 CT: an earlier draft line identified κ as $\cos(17.5°) = 0.953717$ — the corpus writes 0.953939

over 650 times and the digits of $\cos 17.5°$ appear in no file (Claude, corpus grep), so the cosine identification fails at the fourth decimal and is withdrawn. κ's angle is $\arccos(0.953939) = 17.46°$. The exact algebraic angle $\arcsin(1/2\phi) = 18.0000°$ (Appendix A) sits half a degree off. The Cabibbo near-miss is recorded as a miss (6.8σ), not a hit; that standard applies everywhere below. No 45° claim is made (the θ_{23} octant is unresolved).

$E_8 \times E_8$ **clarification (RY, 2026-11-01).** Throughout this paper, "E_8" is RY's shorthand for the double $E_8 \times E_8$ **heterotic structure** (496 dimensions), not the single E_8 (248). Two 248-dimensional sectors face each other across the 4D spacetime boundary interface: the visible sector projects into the Standard Model and local baryonic matter; the shadow sector couples purely through gravity and spacetime curvature. The "Shadow" in Shadow Ladder names this relation — 4D spacetime as the boundary between the sectors. All computed deliverables (240 roots, H_4 folding, Kepler-cube shadow, 10 caustic coordinates, P1/P2) are the boundary-projection geometry of the visible sector and stand unchanged.

The no-go the program must answer: Distler–Garibaldi (2010) — no chiral SM+gravity embedding in E_8 *as a gauge group*. The program's answer, to be demonstrated rather than asserted: the theorem's hypotheses engage E_8-as-gauge-group; the ladder uses E_8 as geometric substrate (root system, lattice, scaffold). Cf. Manogue–Dray–Wilson (2022) on the compactness hypothesis. Open mines, named: chirality, the three-generations problem (Candelas–Horowitz–Strominger–Witten gave four), and the de Sitter asterisk on holography (proven in AdS; our universe accelerates like dS).

2 Geometry closed 2026-10-12

The hexagonal seed, resolved [P]. Brute-forcing all 38,760 six-vertex subsets of the dodecahedron found no hexagon *in* the vertices — that zero stands (re-verified from scratch 2026-10-13). The hexagon lives one level up: the dodecahedron's 20 vertices contain exactly 5 inscribed cubes (125,970 eight-vertex subsets brute-forced; each vertex sits in exactly 2 cubes), and a cube projected along its body diagonal casts a mathematically exact regular hexagon — all gaps 60.0000° (max deviation 2.13×10^{-14} degrees). The hexagon was never hiding in the vertices; it is the **shadow of the inscribed Kepler cube**. The chain closes through the hypercube [D, crew 2026-10-12]: tesseract → rhombic dodecahedron (congruent to canonical coordinates, identical distance multisets) → hexagonal A_2 lattice. Verification script: `artifacts/2026-10-13__muse__shadow-ladder-geometry/dodecahedron_kepler_cube_hexagon.py` in The Hood (filed 2026-10-13 audit; the draft's earlier filing claim is now true).

The $E_8 \to H_4$ angle spectrum [P, with the framing corrected 2026-10-13]. Correction to the draft's earlier wording: E_8's *own* 240 roots have pairwise angles $\{60°, 90°, 120°\}$ only (verified from scratch 2026-10-13; simply-laced, no golden angles). The golden-ratio angles enter through the **Moody–Patera folding** of the E_8 root system onto the H_4 root system — the 240 roots fold 2-to-1 onto the 120 vertices of the 600-cell in $\mathbb{R}^4$ — whose 4D pairwise angle spectrum is exactly $\{0°, 36°, 60°, 72°, 90°, 108°, 120°, 144°, 180°\}$ (crew computation, Antigravity CLI, filed in The Hood). H_4 is not a "maximal subgroup" of the Lie group E_8; it is the Coxeter group obtained by folding. The 18° invariant is **exact**, not approximate: $\arcsin(1/(2\phi)) = \arcsin((\sqrt{5}-1)/4) = 18°$ algebraically (Appendix A). Framing discipline, per the paper's own rules: 18° is decagonal/pentagonal H_2 geometry — $\sin 18° = 1/2\phi$, forced by the golden ratio; H_2 sits inside H_4 (600-cell symmetry), and H_4 is reached from E_8 by folding. The paper does not claim "E_8 predicts 18°" without naming the invariant; it claims the chain $H_2 \subset H_4 \leftarrow E_8$ (folding) forces the angle algebraically.

3 Pre-registered predictions for Euclid

P1 — Lensing dipole asymmetry [D, audited 2026-10-13]. Using the dipole-removed Model B ($|p| = 0.638$ toward $l = 120.7°$, $b = -2.3°$), the crew simulated convergence and shear fields across Euclid's full footprint. Audited against the filed metrics (`artifacts/2026-10-12__antigravity_cli__euclid-roman-caustic-lensing-tomography-and-e8-h/euclid_roman_caustic_lensing_metrics.json`): mean convergence $\langle\kappa\rangle = 0.0576$; dipole asymmetry $\Delta\kappa/\langle\kappa\rangle = 12.34\%$ (aligned vs anti-aligned hemisphere); forecast 123.9σ at 2,500 deg^2 and 293.2σ at 14,000 deg^2 [provisional forecast — see §8] — the two S/N figures scale as $\sqrt{\text{area}}$ (ratio $2.366 = \sqrt{14000/2500}$, verified), so they are one consistent forecast, not two. The 123.9σ figure was computed for the pre-split DR1 definition (2,500 deg^2 *with* weak-lensing products); under ESA's split schedule (§5) the shear test it describes lands with the WL-capable releases. The prediction is binary: the data either decisively detect this asymmetry or the model dies by its own gate. There is no middle ground. (Model A, dipole-inclusive, gives 25.4% / 274.1σ / 648.5σ [provisional forecast] and 8 vertices — quoted for completeness; the paper bets on Model B.)

P2 — Ten caustic vertex targets (sky coordinates, pre-registered).

Target	l (°)	b (°)
V1	257.4	−25.6
V2	257.4	+27.6
V3	173.4	−65.3
V4	173.4	+67.2
V5	308.7	−20.9
V6	308.7	+22.7
V7	1.9	−31.4
V10	266.7	−58.9
V16	321.2	−46.1
V18	49.9	+37.1

Coordinate frame note: galactic (l, b), transcribed from the filed metrics JSON and verified coordinate-by-coordinate 2026-10-13 — all ten match to 0.1°. *Equatorial J2000 equivalents are tabulated in Appendix C.*

Exactly half of all twenty dodecahedron vertices fall inside the unmasked Euclid Wide footprint (10/20 = 50%, per the filed metrics). One further vertex (V3) falls inside the Roman HLS footprint.

P3 — Shape fits, published side by side. Dodecahedron fit residual **6.48°**; hexagonal-prism fit residual **8.71°**. The dodecahedron wins on residual, but both are published and DR1 judges — the prism is not hidden because it lost.

4 Kill criteria (explicit)

- **Triangulation gate: FAILED.** The dipole-removed null (1,000 trials) gave $p = 0.023$ — beats 95% but **fails** the pre-registered $p < 0.01$ kill gate. It does not survive look-elsewhere correction either (∼12 effective trials → $p \approx 0.28$). The best-fit offset sits 124° from the kinematic dipole, so the contamination critique is answered — but the verdict is *suggestive, not significant*, and it is recorded as a failure, not a near-win.
- **DR1 isotropy kills the framework.** If full-DR1 shear maps show unbroken isotropy at the forecast sensitivity, P1 is dead and the ladder's Euclid bet with it. This is the pinch point the paper is named for.

- **Cabibbo stands as a miss.** The 6.8σ near-miss is not re-litigated here.
- **No post-hoc moves.** Predictions pre-date the data by construction (Block timestamp 2026-10-12; Zenodo DOI timestamp on publication). Any prediction added after DR1-Foundation's November 2026 release is labeled post-hoc, not pre-registered.

5 Timing note — read this before citing "November"

ESA split DR1 in two (update published 15 June 2026, `https://www.cosmos.esa.int/web/euclid/dr1-timeline`): **DR1-Foundation (12 November 2026)** releases raw data, calibrated images, catalogues, and spectra over ∼1,900 deg^2 — *no cosmology-derived products*; the **full DR1 (mid-2027)** adds the higher-level galaxy-clustering and weak-lensing products. The decisive shear-map test of P1 therefore lands **mid-2027, not November 2026**. DR1-Foundation catalogues may permit early spot-checks of the §3 vertex targets, but the shear-map test of P1 needs weak-lensing products: the 123.9σ forecast applies to a ∼2,500 deg^2 WL-capable release and the 293.2σ forecast to the full 14,000 deg^2 survey. (This corrects the crew's working shorthand of a "November appointment" — the appointment that decides the model is the WL release.)

6 The race

A Dual Sheet Model researcher pre-registered Euclid DR1 predictions in June 2026 (Hodgkins, Medium). Those predictions are isotropic background quantities ($S_8 \in 0.79$–0.81, w_0/w_a phenomenology) — an orthogonal bet to P1's *directional* anisotropy and P2's sky coordinates. The two programs can both be checked against the same sky without interfering. The reason to publish before DR1-Foundation is the timestamp, not the telescope.

7 What remains open (the honest ledger)

Chirality from a geometric E_8; why three generations; the de Sitter asterisk; an explicit demonstration — not assertion — that the geometric framing sits outside Distler–Garibaldi's hypotheses. Closed in the 2026-10-13 audit: the lensing metrics ($\langle\kappa\rangle$, 12.34%) audited against the filed JSON — both match; $123.9\sigma/293.2\sigma$ scaling verified exact; the S/N figures carry the [provisional forecast] tag (§8); all ten vertex coordinates verified against the filed catalog; the geometry script filed in The Hood; the E_8 angle-spectrum wording corrected (golden angles enter via folding, not from E_8's own roots). The geometry is proven beautiful; §4–§5 are how it proves itself useful.

8 Pipeline caveats (from the 2026-10-14 KILL review)

On 2026-10-14 the crew's adversarial review (Claude Code CLI; the verdict stood under Muse's meta-review) returned five findings against the computational pipeline behind P1/P2. The numbers are not recomputed here — they were pre-registered 2026-10-12 (BLK-119; metrics JSON the same day), and re-running the pipeline now and swapping in new numbers would be the post-hoc move §4 forbids. For a pre-registration, full disclosure of the pipeline's limitations is the honest repair; the mid-2027 weak-lensing test (§5) is the judge. A pipeline correction with every change documented may follow later; there will never be a silent edit.

- **K1 — mapper defaults.** The Model B computation and the 10-target vertex selection were run with the mapper's default parameter set. The defaults are disclosed here as defaults; the

sensitivity of the 10-target list to them is uncharacterized. A sensitivity sweep is explicitly deferred to the post-DR1 correction. The pre-registered list stands frozen.

- **K2 — hardcoded shear factor.** The forecast S/N figures (123.9σ / 293.2σ, and Model A's 274.1σ / 648.5σ) use a hardcoded shear-response factor entering the S/N computation. It is named here as a pipeline constant with stated provenance in the filed pipeline record — not as a derived quantity.
- **K3 — unweighted averages.** $\langle\kappa\rangle = 0.0576$ and $\Delta\kappa/\langle\kappa\rangle = 12.34\%$ are unweighted footprint averages. A weighted recompute is deferred and stated, not hidden.
- **K4 — failed gate.** Disposed by §4 as written: the triangulation gate is recorded FAILED ($p = 0.023$ vs the pre-registered $p < 0.01$ gate; $p \approx 0.28$ after look-elsewhere correction). No change needed.
- **K5 — tier inflation.** The S/N figures are demoted from [D] to [provisional forecast] wherever they appear. Every other tier tag was re-audited against what was actually checked: [P] marks results verified from scratch (hexagon geometry, the 18° identity, the E_8 angle spectrum); [D] marks crew computations audited against filed artifacts (12.34%, $\langle\kappa\rangle$, the ten vertex coordinates).

Recommendation carried to the author's G2 review: keep the S/N figures with this disclosure. Cutting pre-registered numbers from the archival freeze would itself be a post-hoc selection. (That call is the author's.)

A $\arcsin(1/2\phi) = 18°$, one-line proof

$2\phi = 1 + \sqrt{5}$, so $1/(2\phi) = 1/(1+\sqrt{5}) = (\sqrt{5}-1)/4 = \sin 18°$. ■ (Verified to 50-digit arithmetic, 2026-10-12; float diff 5.6×10^{-17}.) Companions: $\cos 36° = \phi/2$; $\sin 18° = \cos 72° = 1/(2\phi)$. Gap to the 17.5° working hypothesis: 0.5° exactly.

κ **addendum (2026-10-13; corrected 15:30 CT):** $\kappa = 0.9539$ is the corpus value $\sqrt{0.91} = 0.953939 = \sqrt{\theta(2-\theta)}$ at $\theta = 7/10$, the χ of the two-channel "at least one" window (chain + mirror; Lean `or_window_mass`). An earlier form of this addendum identified κ as $\cos(17.5°) = 0.953717$ — withdrawn: the corpus writes 0.953939 in 650+ places and the digits of $\cos 17.5°$ in none, so the cosine equality fails at the fourth decimal (Claude, corpus grep 2026-10-13). κ's angle is 17.46°. The 17.5° hypothesis is therefore anchored by κ at 0.04° and by the exact 18° algebraic angle at 0.5°.

B References

1. Kepler, J. *Harmonices Mundi*, Book V (1619) — dodecahedron/cube inscription.
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6. Lisi, A. G. "An Exceptionally Simple Theory of Everything." arXiv:0711.0770 (2007); reply arXiv:1006.4908.

7. Moody, R. V.; Patera, J. — $E_8 \to H_4$ folding (discrete Coxeter reduction, not a Lie subgroup).

8. Hodgkins, M. "Dual Sheet Model DSM Pre-Registration: Predictions for Euclid DR1." Medium, June 2026.

9. Crew working log: The Block, BLK-119 (Muse, 2026-10-12; computations in reply by Antigravity CLI); verification receipts and scripts filed in The Hood.

C Equatorial coordinates of the §3 targets (J2000)

Galactic (l, b) from BLK-119 converted to equatorial (RA, Dec) J2000 with the Murray (1989) FK5→Galactic rotation matrix (the same matrix astropy's Galactic frame uses). Validated: LMC (280.4652, −32.8884) → `05h23m34.60s -69d45′21.95"` (catalog: `05h23m34.6s -69d45′22"`); galactic center and NGP reproduce their IAU definitions to the arcsecond. Only the 10 BLK-119 rows present in the draft are converted — the 7 unfilled rows are not fabricated.

Caveat: cross-checked 2026-10-14 against an independent implementation of the Murray (1989) matrix; all ten coordinates agree to $\leq 0.2''$ *(input precision* $0.1^\circ = 360''$*, so the arithmetic is verified far below the input floor). LMC, galactic-center, and NGP anchors reproduce their reference definitions.*

Target	l (°)	b (°)	RA (J2000)	Dec (J2000)	RA (deg)	Dec (deg)
V1	257.4	−25.6	`06h 17m 19.97s`	`-49d 23′ 59.27"`	94.3332	−49.3998
V2	257.4	+27.6	`10h 02m 17.90s`	`-19d 57′ 10.02"`	150.5746	−19.9528
V3	173.4	−65.3	`02h 07m 54.26s`	`-10d 13′ 37.46"`	31.9761	−10.2271
V4	173.4	+67.2	`11h 19m 53.27s`	`+39d 46′ 57.33"`	169.9719	+39.7826
V5	308.7	−20.9	`15h 37m 12.79s`	`-81d 50′ 41.99"`	234.3033	−81.8450
V6	308.7	+22.7	`13h 19m 11.29s`	`-39d 51′ 38.70"`	199.7970	−39.8607
V7	1.9	−31.4	`20h 10m 33.85s`	`-38d 59′ 23.63"`	302.6410	−38.9899
V10	266.7	−58.9	`02h 43m 47.22s`	`-49d 34′ 31.17"`	40.9468	−49.5753
V16	321.2	−46.1	`22h 41m 23.65s`	`-66d 08′ 44.27"`	340.3486	−66.1456
V18	49.9	+37.1	`16h 53m 38.53s`	`+28d 49′ 47.90"`	253.4105	+28.8300

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Related identifiers: link the Zenodo concept DOIs of the Observerse/E_8-branching papers as "is supplemented by"; link BLK-119 as related.

Canon: this paper is the pre-registration freeze, one canon. Post-DR1 results go in a corrected paper — never edited in place; the old canon is retained as a superseded record.